Introduction to Computer Control Systems, 5 credits, 1RT485

Date: 2024-03-14

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Number of problems: 5

Allowed aid: A non-programmable calculator and mathematical handbooks

Preliminary grades:	grade 3	15 points
	grade 4	21 points
	grade 5	26 points

Some general instructions and information:

- Your solutions should be given in English.
- Write only on one side of the paper.
- Write your exam code and page number on all pages.
- Do not use a red pen.
- Use separate sheets of paper for the different problems (i.e. the numbered problems, 1–5).

With the exception of Problem 1, ***all your answers must be clearly motivated!*** *A correct but unmotivated answer will score reduced points!*

Best of luck!

Formula sheet

Laplace transform table

Table 1: Basic Laplace transforms

$f(t)$	$F(s)$	$f(t)$	$F(s)$
unit impulse — imp(t)	1	$e^{-at}\sin(bt)$	$\frac{b}{(s+a)^2+b^2}$
unit step — step(t)	$\frac{1}{s}$	$e^{-at}\cos(bt)$	$\frac{s+a}{(s+a)^2+b^2}$
unit ramp — ramp(t)	$\frac{1}{s^2}$	$\sinh(bt)$	$\frac{b}{s^2-b^2}$
t	$\frac{1}{s^2}$	$\cosh(bt)$	$\frac{s}{s^2-b^2}$
t^n	$\frac{n!}{s^{n+1}}$	$\frac{1}{2b}t\sin(bt)$	$\frac{s}{(s^2+b^2)^2}$
e^{-at}	$\frac{1}{s+a}$	$t\cos(bt)$	$\frac{s^2-b^2}{(s^2+b^2)^2}$
$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$	$\frac{\cos(bt)-\cos(at)}{a^2-b^2}; (a^2 \neq b^2)$	$\frac{s}{(s^2+a^2)(s^2+b^2)}$
$\frac{1}{(n-1)!}t^{n-1}e^{-at}; (n=1,2,3...)$	$\frac{1}{(s+a)^n}$	$\frac{\sin(at)+at\cos(at)}{2a}$	$\frac{s^2}{(s^2+a^2)^2}$
$\sin(bt)$	$\frac{b}{s^2+b^2}$		
$\cos(bt)$	$\frac{s}{s^2+b^2}$		

Table 2: Properties of Laplace Transforms

$\mathcal{L}[af(t)] = aF(s)$	$\mathcal{L}[tf(t)] = -\frac{dF(s)}{ds}$
$\mathcal{L}[f_1(t)+f_2(t)] = F_1(s)+F_2(s)$	$\mathcal{L}\left[t^2 f(t)\right] = \frac{d^2}{ds^2}F(s)$
$\mathcal{L}\left[\frac{d}{dt}f(t)\right] = sF(s)-f(0)$	$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n}{ds^n}F(s), \quad n=1,2,3,\ldots$
$\mathcal{L}\left[\frac{d^2}{dt^2}f(t)\right] = s^2F(s)-sf(0)-f'(0)$	$\mathcal{L}\left[f\left(\frac{t}{a}\right)\right] = aF(as)$
$\mathcal{L}\left[\int f(t)\,dt\right] = \frac{F(s)}{s}+\frac{1}{s}\left[\int f(t)\,dt\right]_{t=0}$	$\mathcal{L}\left[\int_0^t f_1(t-\tau)f(t)d\tau\right] = F_1(s)F_2(s)$
$\mathcal{L}[f(t-a)] = e^{-as}F(s)$	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$

Open-loop and sensitivity functions

$$G_o(s) = G(s)F_y(s), \qquad S(s) = \frac{1}{1+G_o(s)}, \qquad T(s) = 1-S(s)$$

State-space forms and transfer function relations

- State-space form and transfer function

$$\begin{aligned} \dot{x} &= Ax+Bu \\ y &= Cx+Du \end{aligned} \quad \Rightarrow \quad \boxed{G(s) = C(sI-A)^{-1}B+D}$$

- Associated matrices

$$\mathcal{S} = \begin{bmatrix} B & AB & \cdots & A^{n-1}B \end{bmatrix} \qquad \mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix}$$

- LTI system with transfer function

$$\boxed{G(s) = \frac{b_0 s^n + b_1 s^{n-1} + \cdots + b_n}{s^n + a_1 s^{n-1} + \cdots + a_n}}$$

  i) Observable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 \\ -a_2 & 0 & 1 & \cdots & 0 \\ -a_3 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -a_n & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} b_1 - a_1 b_0 \\ b_2 - a_2 b_0 \\ b_3 - a_3 b_0 \\ \vdots \\ b_n - a_n b_0 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

  ii) Controllable canonical form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} -a_1 & -a_2 & -a_3 & \cdots & -a_n \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = \begin{bmatrix} b_1 - a_1 b_0 & b_2 - a_2 b_0 & \cdots & b_n - a_n b_0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} + b_0 u$$

- Solution to state-space equation

$$\dot{x}(t) = Ax(t) + Bu(t), \quad x(0) = x_0$$

  can be written as

$$\boxed{x(t) = e^{At} x_0 + \int_0^t e^{A\tau} B u(t-\tau) d\tau}$$

- Observer system

$$\boxed{\dot{\hat{x}} = A\hat{x} + Bu + K(y - C\hat{x})}$$

Feedback control structures

General linear feedback in Laplace form:

$$\boxed{U(s) = F_r(s)R(s) - F_y(s)Y(s)}$$

Common control structures in this form.

- PID controller:

$$F_y(s) = F_r(s) = F(s) = K_p + \frac{K_i}{s} + K_d s,$$

  where $K_p, K_i, K_d \geq 0$

- State-feedback controller with observer:

$$\begin{aligned} F_r(s) &= \left(1 - L(sI - A + KC + BL)^{-1}B\right)\ell_0 \\ F_y(s) &= L(sI - A + KC + BL)^{-1}K \end{aligned}$$

Bode's phase diagram

Consider the loop gain $G_o(s) = \mu \frac{\prod_{k=1}^{n_z}(1-\tau_k s)}{s^g \prod_{k=1}^{n_p}(1-T_k s)}$, where for simplicity $\mu > 0$. The phase of $G(i\omega)$ at a given frequency is

$$\arg\left(G_o(i\omega)\right) = -g\,90^\circ + \left[\sum_{k=1}^{n_z} \arctan\left(\tau_k \omega\right) - \sum_{k=1}^{n_p} \arctan\left(T_k \omega\right)\right] \frac{180^\circ}{\pi}$$

Routh–Hurwitz stability criterion

Let $\Phi(s) = a_0 s^n + b_0 s^{n-1} + a_1 s^{n-2} + ...$ be the characteristic polynomial.

$$\begin{array}{ccc} a_0 & a_1 & ... \\ b_0 & b_1 & ... \\ \hline c_0 & c_1 & ... \\ \vdots & \ddots & \end{array}$$

A generic element of the third row, c_k, can be computed as

$$c_k = \frac{b_0 a_{k+1} - a_0 b_{k+1}}{b_0}$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$\boxed{\begin{aligned} x_k &= Fx_k + Gu_k \\ y_k &= Hx_k \end{aligned}}$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Discrete-time state-space forms

A continuous time system with zero-order-hold input signal and sample period T can be written in discrete-time as:

$$
\boxed{\begin{aligned} x(k+1) &= Fx(k) + Gu(k) \\ y(k) &= Hx(k) \end{aligned}}
$$

where

$$
\begin{aligned}
F &= e^{AT} \\
G &= \int_{\tau=0}^{T} e^{A\tau} d\tau B = \lceil \text{if } A^{-1} \text{ exists} \rceil = A^{-1}(e^{AT} - I)B \\
H &= C
\end{aligned}
$$

Bilinear transform formula ($\alpha \in \{0, \frac{1}{2}, 1\}$)

$$
s = \frac{1}{T} \frac{1 - q^{-1}}{\alpha + (1 - \alpha)q^{-1}}
$$

Problem 1: basic questions (6/30)

Answer only *'true'* or *'false'*. Each correct answer gives 1 point, each wrong answer gives -1 point. Minimum total points for Part A and B is 0 , respectively.

Part A

Note: Write 'skip' if your total home assignment score ≥ 8

i) The system

$$G(s) = \frac{(s+5)(s+3)}{(s^2+s-2)(s+4)}$$

is input-output stable.

ii) We control the following system

$$G(s) = \frac{1}{s-4}$$

with a P-controller with constant K. The closed-loop system is unstable for all $K \geq 2$.

iii) The following system is controllable

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

(3 p)

Part B

Note: Write 'skip' if your total home assignment score ≥ 13

i) The rise time and the static gain both relate to the quickness of the system.

ii) For a good controller we wish that, at low frequencies, $|G_c(i\omega)|_{\text{dB}} \approx 0\text{dB}$.

iii) It is desirable for the poles of the state observer to be slower than the poles of the closed-loop system.

(3 p)

Proposed solution to problem 1

Part A

i) False

ii) False

iii) True

Part B

i) False

ii) True

iii) False

Problem 2 (6/30)

a) Consider a system with input $u(t)$ and output $y(t)$ that can be written as

$$Y(s) = G(s)U(s) = \frac{s+1}{s+a}U(s).$$

We want to apply a feedback controller to follow a reference signal $r(t)$. First, consider a P-controller $F(s) = K$.

Given a gain $K > 0$, for which values of a will the closed-loop system $G_c(s)$ be stable?

(3 p)

b) Consider the system above with $a = 2$ (that is, $G(s) = \frac{s+1}{s+2}$) and a feedback controller $F(s) = 5 + \frac{K_i}{s}$.

Let the reference signal $r(t)$ be a step. That is,

$$r(t) = \begin{cases} r_0, & t \geq 0 \\ 0 & \text{otherwise} \end{cases}.$$

Set $K_i = 1$ and calculate the stationary error of the controller: $e(t) = r(t) - y(t)$ as $t \to \infty$.

(3 p)

Proposed solution to problem 2

a) The closed-loop system is given by

$$G_c(s) = \frac{F(s)G(s)}{1+F(s)G(s)} = \frac{K\frac{s+1}{s+a}}{1+K\frac{s+1}{s+a}} = \frac{K(s+1)}{(1+K)s+a+K}$$

Hence, the closed-loop system has a pole in $s = -\frac{a+K}{1+K}$, which is negative if $a+K>0$ since $K>0 \implies (1+K)>0$.
Therefore, the closed-loop system is stable if $a > -K$.

(3 p)

b) Now, the close-loop system is instead

$$G_c(s) = \frac{F(s)G(s)}{1+F(s)G(s)} = \frac{(5+\frac{1}{s})\frac{s+1}{s+2}}{1+(5+\frac{1}{s})\frac{s+1}{s+2}} = \frac{(5+\frac{1}{s})(s+1)}{s+2+(5+\frac{1}{s})(s+1)} = \frac{(5s+1)(s+1)}{s^2+2s+(5s+1)(s+1)}$$

Moreover, we have that

$$Y(s) = G_c(s)R(s) = G_c(s)\frac{r_0}{s}$$

Since $G_c(s)$ is stable, we can apply the final value theorem to get

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} sG_c(s)\frac{r_0}{s} = G_c(0)r_0 = 1\cdot r_0$$

The steady-state error is

$$\lim_{t\to\infty} e(t) = \lim_{t\to\infty} r(t) - y(t) = r_0 - r_0 = 0.$$

(3 p)

Problem 3 (6/30)

You have been tasked with designing a cruise control system for a car. The objective is to maintain a constant speed y when the vehicle is driven on a flat road by applying the right amount of force u via the car's engine.

The dynamics of the system is well approximated by the transfer function:

$$G(s) = \frac{1}{ms + b},$$

where $m > 0$ is the mass of the car and $b > 0$ is a friction term.

According to what you studied in the course, you realize that a good choice for the $F(s)$ controller might be a PI controller with gain $K > 0$:

$$F(s) = K\frac{1 + 10s}{s}$$

a) You first design the controller's gain K based on the *nominal* system that assumes $m = 1000$ and $b = 1$. For what values of K is the closed-loop bandwidth greater than 0.1 rad/s and the phase margin greater than 45°?

(3 p)

b) Suppose that you ended up choosing $K = 20$, resulting in the complementary sensitivity function shown in Figure 1.

You realize, however, that when the control system is applied to the car, the mass could vary up to 200kg depending on the passengers. The real system is therefore:

$$G^o(s) = \frac{1}{(1000 + \delta_m)s + 1},$$

where $-200 \leq \delta_m \leq 200$. Is the closed-loop stability still guaranteed?

Hint: *consider the extreme cases,* $\delta_m = 200$ *and* $\delta_m = -200$. **(3 p)**

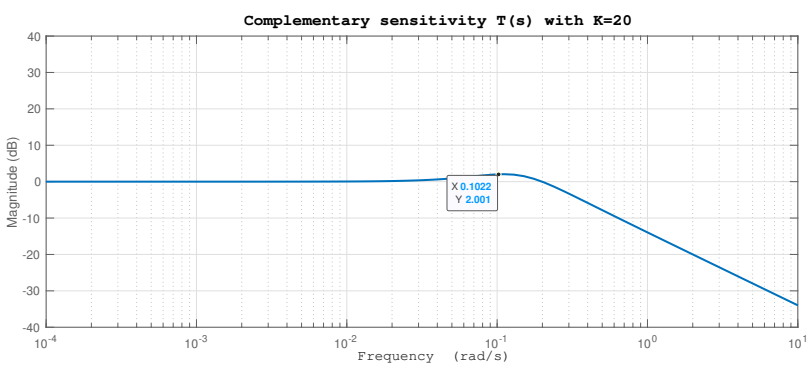


Figure 1: Magnitude plot of the complementary sensitivity function for $K = 20$.

Proposed solution to problem 3

a) Plotting the asymptotic bode diagram of $G_o(s) = \frac{(1+10s)}{s(1+1000s)}$ (i.e., we initially set $K = 1$), we realize that at the desired crossover frequency of 0.1 rad/s the magnitude is $|G_o(i0.1)|_{\text{dB}} = -20\text{dB}$. Therefore, in order to have $\omega_c \geq 0.1$ we must have $K \geq 20\text{dB}$, or equivalently $K \geq 10$.

As for the phase margin requirement, we note that when the crossover frequency is bigger than 0.1, the phase loss due to the two poles is almost $-180°$ and hence

$$\angle G(i\omega_c) \approx -180 + \frac{180}{\pi} \arctan(10\omega_c).$$

The phase margin is $\varphi_m = \frac{180}{\pi} \arctan(10\omega_c)$. Note that if $\omega_c > 0.1$, $\varphi_m \geq 45°$.

b) Let's compute the relative mismatch $\Delta(s)$ such that $G^0(s) = G(s)(1+\Delta(s))$

$$\frac{1}{(1000+\delta_m)s+1} = \frac{1}{1000s+1}(1+\Delta(s)) \quad \rightarrow \quad \Delta(s) = \frac{-\delta_m s}{(1000+\delta_m)s+1}$$

Note that

- $\Delta(s)$ does not change the number of unstable poles of $G(s)$, i.e. 0.
- $\Delta(s)$ is causal, i.e. it does not have more zeros than poles, so $F(i\omega)G^0(i\omega) \rightarrow 0$ as $\omega \rightarrow \infty$.

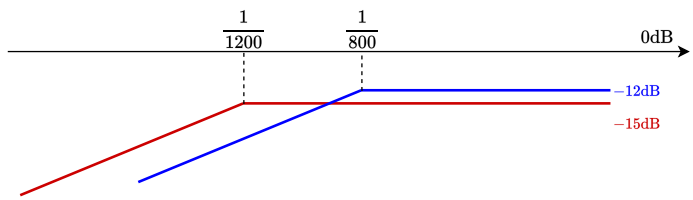
We can apply the Robustness criterion (see Result 6.2 in F7).

Case $\delta_m = 200$: $\quad \Delta(s) = \frac{-200s}{1200s+1}$

At high frequency $|\Delta(i\omega)| = \left|\frac{200}{1200}\right| \approx -15\text{dB}$. The Bode plot of this transfer function is the red line in the figure below. Since $T(i\omega) < 2\text{dB}$, we can note that $|T(i\omega)|_{\text{dB}} < -|\Delta(i\omega)|_{dB}$ at every ω. The robustness condition is satisfied.

Case $\delta_m = -200$**:** $\quad \Delta(s) = \frac{200s}{800s+1}$

At high frequency $|\Delta(i\omega)| = \left|\frac{200}{800}\right| \approx -12\text{dB}$. The Bode plot of this transfer function is the blue line in the figure below. As in the previous case $|T(i\omega)|_{\text{dB}} < -|\Delta(i\omega)|_{dB}$ at every ω, so the robustness condition is satisfied.



<u>Alternative solution</u>: Compute $G_c^o = \frac{F(s)G^o(s)}{F(s)G^o(s)}$ and verify that its poles have negative real part (e.g., using Routh criterion).

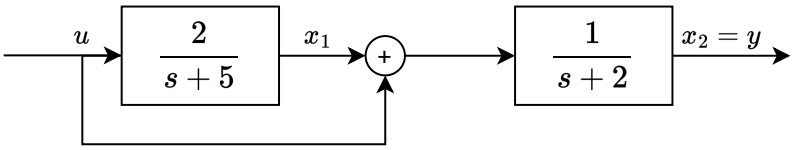


Figure 2: Block diagram of the chemical system

Problem 4 (7/30)

A chemical system can be modeled by the block diagram depicted in the figure above.

a) Derive a state-space representation where the states x_1 and x_2, and the output y, are those reported in the Figure.

(2 p)

Note: *if you cannot solve point a), the following points can be solved using one of the canonical forms.*

b) Using the state-space model you just derived, design a state-feedback control law $u = -Lx + \ell_0 r$ so that the closed-loop eigenvalues are in -7 and -10. Find ℓ_0 so that, for constant references, the tracking error is 0.

(3p)

c) Unfortunately, not all the states are measurable. Design a state observer so that its state estimation error has both the eigenvalues in -20.

(2 p)

Proposed solution to problem 4

a) From the scheme, we can write

$$X_1(s) = \frac{2}{s+5}U(s) \to sX_1(s) = -5X_1(s) + 2U(s) \to \dot{x}_1 = -5x_1 + 2u$$

$$X_2(s) = \frac{1}{s+2}(X_1(s)+U(s)) \to sX_2(s) = -2X_2(s)+U(s)+X_1(s) \to \dot{x}_2 = x_1-2x_2+u$$

Therefore, the state-space form is

$$\begin{aligned} \dot{x} &= \begin{bmatrix} -5 & 0 \\ 1 & -2 \end{bmatrix} x + \begin{bmatrix} 2 \\ 1 \end{bmatrix} u \\ y &= \begin{bmatrix} 0 & 1 \end{bmatrix} x \end{aligned}$$

Note that $\frac{Y(s)}{U(s)} = \left(1 + \frac{2}{s+5}\right)\frac{1}{s+2} = \frac{s+7}{s^2+7s+10}$.

b) First, we verify that the system is controllable:

$$\mathcal{S} = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 2 & -10 \\ 1 & 0 \end{bmatrix} \to \text{ok}$$

Then, we compute the desired characteristic polynomial

$$(\lambda+7)(\lambda+10) = \lambda^2 + 17\lambda + 70$$

The closed-loop characteristic polynomial is

$$\begin{aligned} \varphi(\lambda) = \det(\lambda I - (A - BL)) &= \det \begin{bmatrix} \lambda+5+2\ell_1 & 2\ell_2 \\ -1+\ell_1 & \lambda+2+\ell_2 \end{bmatrix} \\ &= (\lambda+5+2\ell_1)(\lambda+2+\ell_2) - 2\ell_2(\ell_1-1) \\ &= \lambda^2 + (5+2\ell_1+2+\ell_2)\lambda + (5+2\ell_1)(2+\ell_2) - 2\ell_1\ell_2 + 2\ell_2 \end{aligned}$$

Setting $\varphi(\lambda)$ to the desired characteristic polynomial we get

$$\begin{cases} 7+2\ell_1+\ell_2 = 17 \\ 10+7\ell_2+4\ell_1 = 70 \end{cases} \to \begin{cases} \ell_2 = 10-2\ell_1 \\ -10\ell_1 = -10 \end{cases} \to L = \begin{bmatrix} 1 & 8 \end{bmatrix}$$

Finally, we need to compute ℓ_0 so that $G_c(0) = 1$.

$$\begin{aligned} \ell_0 &= \frac{1}{C(-A+BL)^{-1}B} = \frac{1}{\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 7 & 16 \\ 0 & 10 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}} = \frac{70}{\begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & -16 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}} \\ &= \frac{70}{\begin{bmatrix} 0 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix}} = \frac{70}{7} = 10 \end{aligned}$$

c) First, we verify that the system is observable:

$$\mathcal{S} = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix} \to \text{ok}$$

Then, we compute the desired characteristic polynomial

$$(\lambda + 20)(\lambda + 20) = \lambda^2 + 40\lambda + 400$$

The characteristic polynomial of the state estimation error is

$$\begin{aligned} \varphi(\lambda) &= \det(\lambda I - (A - KC)) = \det \begin{bmatrix} \lambda + 5 & \kappa_1 \\ -1 & \lambda + 2 + \kappa_2 \end{bmatrix} \\ &= \lambda^2 + (7 + \kappa_2)\lambda + 10 + 5\kappa_2 + \kappa_1 \end{aligned}$$

Matching the coefficients of this polynomial to those of the desired characteristic polynomial we get

$$\begin{cases} 7 + \kappa_2 = 40 \\ 10 + 5\kappa_2 + \kappa_1 = 400 \end{cases} \quad \to \quad K = \begin{bmatrix} 225 \\ 33 \end{bmatrix}$$

Problem 5 (5/30)

For the system $G(s) = 5\frac{s+7}{(s+5)(s+2)^2}$ a colleague of yours has designed a controller

$$F(s) = 10\frac{s+2}{s+7}$$

This controller achieves a closed-loop bandwidth $\omega_B \approx \omega_c = 6$ rad/s and a phase margin $\varphi_m = 58°$.

a) What is a range of sampling frequencies that could be adopted to sample the error signal? Justify.

(1.5 p)

b) Choose a sampling frequency within the proposed range and discretize, with the corresponding sampling time, the control law $U(s) = F(s)E(s)$ by a method of your choice among Forward Euler, Tustin, and Backward Euler. The discretized control law should be in the form $u_k = a_1 u_{k-1} + b_0 e_k + b_1 e_{k-1}$.

(2.5 p)

c) What is the loss of phase margin due to the discretization of the controller? Is the residual phase margin still acceptable?

(1 p)

Proposed solution to problem 5

a) The sampling frequency is usually chosen in the range $[20\omega_c, 40\omega_c]$ or $[20\omega_c, 50\omega_c]$. In this case, we can take $f_s \in (20, 50)$ Hz. A too small sampling frequency implies that we do not sample enough the error signal, meaning that we incur into aliasing. A too large sampling frequency implies an excessive hardware cost.

b) We pick $f_s = 25$ Hz, hence the sampling time is $T = \frac{1}{25} = 0.04$. Choosing the Tustin discretization, we substitute

$$s = \frac{2}{0.04}\frac{1-q^{-1}}{1+q^{-1}}$$

into $F(s)$. We thus get

$$F(q^{-1}) = 10\frac{50\frac{1-q^{-1}}{1+q^{-1}} + 2}{50\frac{1-q^{-1}}{1+q^{-1}} + 7} = 20\frac{50 - 50q^{-1} + 2 + 2q^{-1}}{50 - 50q^{-1} + 7 + 7q^{-1}} = \frac{520 - 480q^{-1}}{57 - 43q^{-1}}$$

The corresponding control law is

$$u_k = \frac{43}{57}u_{k-1} + \frac{520}{57}e_k - \frac{480}{57}e_{k-1}$$

c) The phase loss is

$$\Delta\varphi_m \approx -\frac{\omega_c}{2\pi f_s}180 \approx -7^\circ.$$

The residual phase $\varphi'_k = 58^\circ - 7^\circ$ is bigger than 45°, which is acceptable.